

AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Workforce Longevity: Designing Data-Led Retention Strategies

Project Domain

Excel / Power BI with AI Integration

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:



Student Name

Anamika Shah



Student Email

annexi.mika@gmail.com



Guide

Ms. Umamaheshwar

Institution:

Graphic Era Hill University

Year - 2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project guide, Ms. Umamaheshwari, for her continuous support, guidance, and valuable feedback throughout the completion of this project. Her insights and encouragement played a crucial role in shaping this work.

I would also like to extend my heartfelt thanks to our instructor, Sagar Sir, for his guidance, support, and assistance during the course of this project. His inputs helped in better understanding and execution of the project work.

I am also thankful to the Edunet Foundation, Microsoft, LinkedIn, and SAP for providing this valuable fellowship opportunity, along with the necessary learning resources and platform to enhance my skills in AI and data analytics.

I extend my gratitude to my institution, Graphic Era Hill University, for providing the required infrastructure and support during the project.

Finally, I would like to thank my family and friends for their constant encouragement and motivation throughout this journey.

ABSTRACT OF THE PROJECT

This project is part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program, a collaborative initiative by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP. The objective of this project is to design a data-driven employee retention framework using HR analytics and Power BI.

Employee attrition is a significant business challenge that affects productivity, increases operational costs, and impacts organizational stability. Although organizations collect large volumes of employee data, it is often not effectively utilized to support decision-making or identify attrition risks.

This project uses the IBM HR Analytics Employee Attrition & Performance dataset obtained from Kaggle to analyze workforce patterns and identify key drivers of attrition such as job satisfaction, overtime, salary, and tenure. Data cleaning and preprocessing were performed using Excel and Power BI to ensure data accuracy and consistency. Key performance indicators (KPIs) such as Attrition Rate, Tenure Distribution, and Risk Segmentation were developed using Power BI DAX.

An interactive dashboard was created to visualize employee trends and provide dynamic insights through filtering and segmentation. The project further translates these insights into actionable retention strategies, enabling organizations to proactively identify high-risk employees and improve workforce planning.

The outcome of this project is not only an analytical dashboard but also a strategic decision-support framework that helps organizations shift from reactive to proactive employee retention practices, ultimately enhancing organizational performance.

TABLE OF CONTENTS

Abstract of the Project

List of Figures

List of Tables

Chapter 1 — Introduction

- 1.1 Problem Statement
- 1.2 Motivation
- 1.3 Objectives
- 1.4 Scope of the Project

Chapter 2 — Literature Survey

Chapter 3 — Proposed Methodology

- 3.1 System Design
- 3.2 Modules / Components Used
- 3.3 Data Flow Diagram (DFD)
- 3.4 Advantages
- 3.5 Requirement Specification

Chapter 4 — Implementation and Results

- 4.1 Data Cleaning Results
- 4.2 Visualization / Output Results
- 4.3 AI Tool Integration Results

Chapter 5 — Discussion and Conclusion

References

Appendices

LIST OF FIGURES

Figure No.	Figure Title / Description	Page No.
Figure 1	Raw IBM HR Dataset Screenshot	16
Figure 2	Cleaned Dataset / Data Validation Summary	16
Figure 3	Executive Overview Dashboard (Page 1)	17
Figure 4	Driver Analysis Dashboard (Page 2)	18
Figure 5	Retention Strategy Dashboard (Page 3)	18

Figure 6	DFD Level 0 – Context Diagram	13
Figure 7	DFD Level 1 – Process Flow Diagram	13

LIST OF TABLES

Table No.	Table Title / Description	Page No.
Table 1	Raw Dataset Summary (Fields and Structure)	15
Table 2	Data Cleaning Summary (Issues Identified and Resolved)	15
Table 3	Hardware Requirements	14
Table 4	Software Requirement	14
Table 5	Key Findings and Insights Summary	21

CHAPTER 1

Introduction

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the objectives and scope of the project. The fellowship projects span multiple domains including Data Analytics, Business Intelligence, Healthcare Analytics, Content Generation, and AI-assisted Reporting.

1.1 Problem Statement

The main problem addressed in this project is employee attrition, which is a major challenge in the domain of Human Resource Analytics. Organizations often struggle to understand why employees leave, despite having access to large volumes of workforce data that remain underutilized.

This problem is significant because high attrition leads to reduced productivity, increased hiring and training costs, and disruption in business operations. It also negatively impacts employee morale and long-term organizational stability, making it a critical issue in real-world business environments.

The primary stakeholders affected by this problem are organizations, HR departments, and management teams responsible for workforce planning and employee retention. Ineffective handling of attrition directly impacts organizational performance, decision-making efficiency, and workforce continuity.

A key limitation of existing approaches is that they are largely reactive rather than predictive. Organizations typically take action only after employees leave, and there is a lack of structured, data-driven systems to analyze workforce patterns, identify early warning signals, and support timely interventions. Additionally, raw HR data is often underutilized due to insufficient analytical frameworks and lack of automation.

Therefore, there is a need for a data-driven and predictive approach that can transform raw employee data into meaningful insights, identify high-risk employees in advance, and support proactive and automated retention strategies.

1.2 Motivation

The motivation behind this project comes from the growing importance of data-driven decision-making in modern organizations. Employee attrition is a critical issue that directly impacts business performance, making it essential to understand and manage effectively.

From a professional perspective, this project provides an opportunity to apply skills in data analytics, Power BI, and AI tools to solve a real-world business problem. It also aligns with career interests in HR analytics and business intelligence.

In real-world applications, this project can help organizations identify high-risk employees, understand key factors influencing attrition, and design effective retention strategies. It enables companies to shift from reactive to proactive workforce management.

The project has the potential to improve workforce planning, reduce employee turnover, and enhance overall organizational efficiency and performance.

Furthermore, this project aligns with current industry trends where organizations increasingly rely on AI, data analytics, and business intelligence tools to gain insights and support strategic decision-making.

1.3 Objectives

1. To collect and understand the structure of the IBM HR Analytics dataset used in the project.
2. To clean and preprocess raw data by removing duplicates, handling missing values, and ensuring data consistency.
3. To perform HR analytics using Excel and Power BI to identify patterns, trends, and key workforce insights.
4. To develop key performance indicators (KPIs) such as Attrition Rate, Tenure Distribution, and Risk Segmentation.
5. To build interactive dashboards that effectively communicate insights and support data-driven decision-making.
6. To apply predictive analytics using Azure ML to estimate employee attrition risk and identify high-risk employees.
7. To integrate workflow automation using Zapier for real-time reporting and automated HR alerts.
8. To design an end-to-end data-driven retention framework for proactive and automated workforce planning.

1.4 Scope of the Project

This project is based on the IBM HR Analytics Employee Attrition & Performance dataset obtained from Kaggle, which includes employee-related information such as demographics, job roles, salary, job satisfaction, and attrition status.

The tools used in this project include Microsoft Excel, Power BI, and AI tools such as ChatGPT and Microsoft Copilot for data analysis and insight generation.

The analysis focuses on data cleaning, preprocessing, KPI development, and interactive dashboard visualization to identify patterns and key drivers of employee attrition.

The project further incorporates predictive modeling using Azure ML to estimate attrition probability and workflow automation using Zapier for near real-time reporting and automated alerts.

This project is developed as part of the AICW Fellowship Program and is intended as a conceptual and analytical solution rather than a full-scale production-level implementation.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research

Employee attrition and workforce analytics have been widely studied in the domain of Human Resource Management and Business Intelligence. Many organizations use traditional data analysis techniques in tools like Microsoft Excel to clean, process, and analyze employee data. Excel-based methods such as pivot tables, filtering, and statistical analysis are commonly used to identify basic patterns in workforce data.

With the advancement of business intelligence tools, platforms such as Microsoft Power BI have become widely adopted for HR analytics. Power BI enables interactive dashboards, real-time visualization, and advanced data modeling, allowing organizations to better understand employee behavior and attrition trends. Several studies highlight the effectiveness of dashboard-based analytics in improving decision-making and organizational performance.

In recent years, the integration of Artificial Intelligence tools such as Microsoft Copilot and AI chatbots has further enhanced data analysis capabilities. These tools assist in generating insights, summarizing data, and automating repetitive analytical tasks, making data-driven decision-making more efficient and accessible.

However, most existing approaches either focus on complex machine learning models or basic reporting dashboards. There is often a lack of structured frameworks that combine data cleaning, analysis, visualization, and strategic decision-making into a single workflow. Additionally, many organizations still follow reactive approaches, where actions are taken after attrition occurs rather than predicting and preventing it.

This project addresses these gaps by developing a data-driven retention framework that integrates data preprocessing, KPI development, and interactive dashboard visualization using Power BI. It focuses on transforming raw HR data into actionable insights and supporting proactive workforce retention strategies.

2.2 Tools and Technologies Review

Microsoft Excel — Microsoft Excel is used for initial data cleaning and preprocessing. It helps in handling missing values, removing duplicates, and performing basic data transformations. Features such as sorting, filtering, and data validation ensure data accuracy and consistency before analysis.

Microsoft Power BI — Power BI is the primary tool used for data analysis and visualization. It enables the creation of interactive dashboards, KPI cards, and dynamic reports. DAX (Data Analysis Expressions) is used to calculate key metrics such as Attrition Rate, Tenure Groups, and Risk Segmentation.

Microsoft Copilot (AI) — Microsoft Copilot assists in generating insights, summaries, and interpretations from data. It helps automate analytical tasks and improves efficiency in understanding complex datasets.

AI Chatbots (ChatGPT) — ChatGPT is used for generating analytical explanations, structuring the report, and assisting in interpreting insights. It enhances productivity by providing quick and accurate content generation and analysis support.

Azure Machine Learning (Azure ML) — Azure ML is used to develop a predictive model that estimates employee attrition probability based on key factors such as job satisfaction, salary, overtime, and tenure. It enables early identification of high-risk employees and supports proactive decision-making.

Zapier — A workflow automation tool used to integrate Google Sheets with Gmail for automated HR reporting and real-time insight delivery.

2.3 Gaps Identified

- Most existing literature focuses on complex machine learning approaches, while simpler, tool-based solutions using Excel and Power BI for non-technical users are relatively underexplored.
- There is limited practical documentation on integrating AI tools such as Microsoft Copilot and ChatGPT with Excel and Power BI for domain-specific HR analytics, especially in student-level or real-world applications.
- Prompt engineering practices for business and analytics use cases are still evolving and lack standardized frameworks, which affects the consistency and reliability of AI-generated insights.
- Additionally, many existing solutions focus only on data analysis or visualization but lack a structured framework that connects data processing, predictive analytics, and strategic decision-making for employee retention.
- There is also a lack of integration between analytics and workflow automation tools, resulting in limited real-time insight delivery and delayed HR decision-making processes.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

The system design of this project follows a structured data analytics workflow, where raw employee data is transformed into meaningful insights through data cleaning, analysis, predictive modeling using Azure ML, and visualization. The overall process ensures a smooth flow from input to actionable output, supported by automated reporting using Zapier.

Data Source / Input: The project uses the IBM HR Analytics Employee Attrition & Performance dataset obtained from Kaggle. The dataset contains employee information such as age, gender, department, job role, monthly salary, job satisfaction, work experience, and attrition status. The data is provided in a structured format and serves as the primary input for analysis.

Data Cleaning: Data cleaning and preprocessing were performed using Microsoft Excel and Power BI. This includes removing duplicate records, handling missing values, correcting data types, and standardizing categorical fields. Additional derived columns such as Salary Bands and Tenure Groups were created to enhance analysis.

Analysis / Processing: The cleaned data was analyzed using Power BI. Key performance indicators (KPIs) such as Attrition Rate, Total Employees, and High-Risk Employee Count were calculated using DAX measures. Data segmentation and filtering techniques were applied to identify patterns, trends, and key drivers of employee attrition.

Predictive Modeling: Azure ML is integrated into the system to develop a predictive model that estimates employee attrition probability based on key factors such as job satisfaction, salary, overtime, and tenure. This enables early identification of high-risk employees and supports proactive decision-making.

Visualization / Output: The final output of the project is an interactive Power BI dashboard consisting of multiple pages, including Executive Overview, Driver Analysis, and Retention Strategy. The dashboard provides visual insights through charts, KPI cards, and filters, enabling users to explore data dynamically and support data-driven decision-making.

Automation / Reporting: Zapier is integrated to automate the reporting process by connecting Google Sheets with email platforms. It triggers automated alerts based on defined conditions, such as high attrition probability, ensuring near real-time insight delivery and reducing manual effort.

3.2 Modules / Components Used

- **Module 1 — Data Ingestion:** The raw IBM HR Analytics dataset is loaded into Microsoft Excel and Power BI for further processing and analysis.

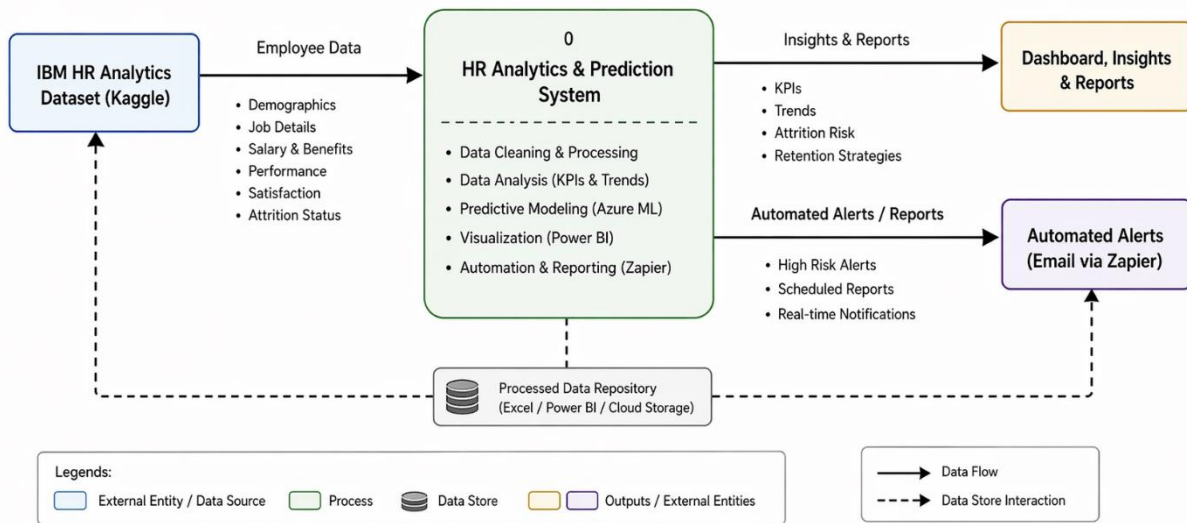
- **Module 2 — Data Cleaning:** This module involves removing duplicate records, handling missing values, correcting data types, and standardizing data formats to ensure accuracy and consistency.
- **Module 3 — Data Analysis:** In this stage, data is analyzed using Power BI and DAX measures to calculate key metrics such as Attrition Rate, Total Employees, and Risk Segmentation. Patterns and trends in employee behavior are identified.
- **Module 4 — Predictive Modeling:** This module uses Azure ML to develop a predictive model that estimates employee attrition probability based on key factors such as job satisfaction, salary, overtime, and tenure. It enables early identification of high-risk employees and supports proactive decision-making.
- **Module 5 — Visualization / Report Generation:** Interactive dashboards are created using Power BI, including charts, KPI cards, and slicers to present insights in a clear and user-friendly manner.
- **Module 6 — AI Integration:** AI tools such as ChatGPT or Microsoft Copilot are used to generate insights, summaries, and assist in interpreting data, enhancing the overall analysis process.
- **Module 7 — Retention Strategy Framework:** This module focuses on translating analytical and predictive insights into actionable retention strategies, identifying high-risk employees, and supporting proactive workforce planning.
- **Module 8 — Automation & Reporting:** This module integrates Zapier to automate HR reporting by connecting Google Sheets with email platforms. It enables near real-time data flow and automatic alerts based on defined conditions, reducing manual effort and improving reporting efficiency.

3.3 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data flows through your system. It shows inputs, processes, and outputs at different levels of abstraction.

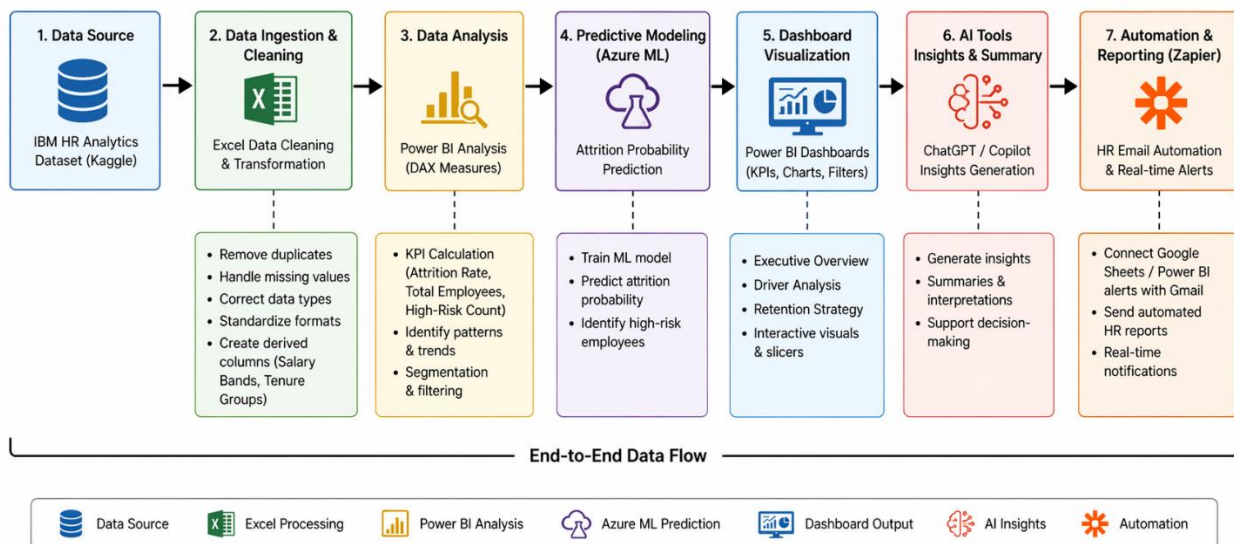
DFD Level 0 — Context Diagram

DFD Level 0 – Context Diagram



DFD Level 1 — Detailed Process Flow

HR Analytics – Employee Attrition Analysis System



3.4 Advantages

The proposed approach offers several advantages in terms of efficiency, usability, and decision-making:

- Automates data cleaning and preprocessing, reducing manual effort and improving data accuracy

- Provides clear and interactive visual insights that are easy to understand for non-technical stakeholders
- Leverages AI tools such as ChatGPT and Copilot to accelerate analysis and enhance insight generation
- Enables proactive decision-making by identifying high-risk employees using predictive modeling (Azure ML)
- Supports early detection of attrition risk, allowing timely and preventive HR interventions
- Scalable and adaptable, as the same framework can be applied to larger datasets and different business domains
- Cost-effective solution, utilizing widely available tools like Excel, Power BI, and cloud-based AI services
- Enables automated HR reporting using Zapier, ensuring near real-time insight delivery and reducing manual effort
- Integrates analytics, prediction, and automation into a unified end-to-end HR decision-support system

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot / Chatbot access

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

1. The dataset used in this project is the IBM HR Analytics Employee Attrition & Performance dataset, which was downloaded from Kaggle. The dataset contains detailed employee information such as demographics, job roles, salary, job satisfaction, and attrition status.

Table 1: Raw Dataset Summary (Fields and Structure)

Field	Description
Age	Employee age
Department	Department name
Job Role	Role of employee
Monthly Income	Salary of employee
Attrition	Employee status (Yes/No)

2. The original dataset consists of approximately 1470 rows and 35 columns, representing various attributes related to employees.
3. During initial exploration, several issues were identified in the dataset, including inconsistent categorical values (such as underscores in travel frequency), presence of unnecessary columns, and the need for better data structuring for analysis. Although the dataset was relatively clean, minor preprocessing was required to improve usability.
4. The following data cleaning steps were applied:
 - Removal of unnecessary columns such as Employee Count and Employee Number
 - Standardization of categorical values (e.g., replacing underscores with spaces in Travel Frequency)
 - Renaming column headers for better clarity and understanding
 - Creation of derived columns such as Salary Band and Tenure Group
 - Validation of data types and consistency across all fields.

Table 2: Data Cleaning Summary (Issues Identified and Resolved)

Issue Identified	Action Taken
Inconsistent categorical values	Standardized formats
Unnecessary columns	Removed
Formatting issues	Cleaned and structured
Missing values	Verified and handled

- After cleaning and preprocessing, the dataset structure was optimized for analysis, while maintaining the same number of rows. The cleaned dataset is more structured, consistent, and suitable for further analysis and visualization in Power BI.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Age	Attrition	BusinessTravel	DailyRate	Department	Distance	Education	EducationField	Employee	Employee	Environm	Gender	HourlyRat	JobInvolv	JobLevel	JobRole
2	41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	1	1	2	Female	94	3	2	Sales Exec
3	49	No	Travel_Frequently	279	Research & Development	8	1	Life Sciences	1	2	3	Male	61	2	2	Research
4	37	Yes	Travel_Rarely	1373	Research & Development	2	2	Other	1	4	4	Male	92	2	1	Laborator
5	33	No	Travel_Frequently	1392	Research & Development	3	4	Life Sciences	1	5	4	Female	56	3	1	Research
6	27	No	Travel_Rarely	591	Research & Development	2	1	Medical	1	7	1	Male	40	3	1	Laborator
7	32	No	Travel_Frequently	1005	Research & Development	2	2	Life Sciences	1	8	4	Male	79	3	1	Laborator
8	59	No	Travel_Rarely	1324	Research & Development	3	3	Medical	1	10	3	Female	81	4	1	Laborator
9	30	No	Travel_Rarely	1358	Research & Development	24	1	Life Sciences	1	11	4	Male	67	3	1	Laborator
10	38	No	Travel_Frequently	216	Research & Development	23	3	Life Sciences	1	12	4	Male	44	2	3	Manufact
11	36	No	Travel_Rarely	1299	Research & Development	27	3	Medical	1	13	3	Male	94	3	2	Healthcar
12	35	No	Travel_Rarely	809	Research & Development	16	3	Medical	1	14	1	Male	84	4	1	Laborator
13	29	No	Travel_Rarely	153	Research & Development	15	2	Life Sciences	1	15	4	Female	49	2	2	Laborator
14	31	No	Travel_Rarely	670	Research & Development	26	1	Life Sciences	1	16	1	Male	31	3	1	Research
15	34	No	Travel_Rarely	1346	Research & Development	19	2	Medical	1	18	2	Male	93	3	1	Laborator
16	28	Yes	Travel_Rarely	103	Research & Development	24	3	Life Sciences	1	19	3	Male	50	2	1	Laborator
17	29	No	Travel_Rarely	1389	Research & Development	21	4	Life Sciences	1	20	2	Female	51	4	3	Manufact
18	32	No	Travel_Rarely	334	Research & Development	5	2	Life Sciences	1	21	1	Male	80	4	1	Research
19	22	No	Non-Travel	1123	Research & Development	16	2	Medical	1	22	4	Male	96	4	1	Laborator
20	53	No	Travel_Rarely	1219	Sales	2	4	Life Sciences	1	23	1	Female	78	2	4	Manager

Raw (uncleaned) IBM HR dataset

	A	B	C	D	E	F	G	H	I
1	Age	Attrition Status	Travel Frequency	Daily Pay Rate	Department	Commute Distance	Education Level	Field of Education	Work Environment Satisfaction
2	41	Yes	Travel Rarely	1102	Sales	1	2	Life Sciences	2
3	49	No	Travel Frequently	279	Research & Development	8	1	Life Sciences	3
4	37	Yes	Travel Rarely	1373	Research & Development	2	2	Other	4
5	33	No	Travel Frequently	1392	Research & Development	3	4	Life Sciences	4
6	27	No	Travel Rarely	591	Research & Development	2	1	Medical	1
7	32	No	Travel Frequently	1005	Research & Development	2	2	Life Sciences	4
8	59	No	Travel Rarely	1324	Research & Development	3	3	Medical	3
9	30	No	Travel Rarely	1358	Research & Development	24	1	Life Sciences	4
10	38	No	Travel Frequently	216	Research & Development	23	3	Life Sciences	4
11	36	No	Travel Rarely	1299	Research & Development	27	3	Medical	3
12	35	No	Travel Rarely	809	Research & Development	16	3	Medical	1
13	29	No	Travel Rarely	153	Research & Development	15	2	Life Sciences	4
14	31	No	Travel Rarely	670	Research & Development	26	1	Life Sciences	1
15	34	No	Travel Rarely	1346	Research & Development	19	2	Medical	2
16	28	Yes	Travel Rarely	103	Research & Development	24	3	Life Sciences	3
17	29	No	Travel Rarely	1389	Research & Development	21	4	Life Sciences	2
18	32	No	Travel Rarely	334	Research & Development	5	2	Life Sciences	1
19	22	No	Non-Travel	1123	Research & Development	16	2	Medical	4
20	53	No	Travel Rarely	1219	Sales	2	4	Life Sciences	1

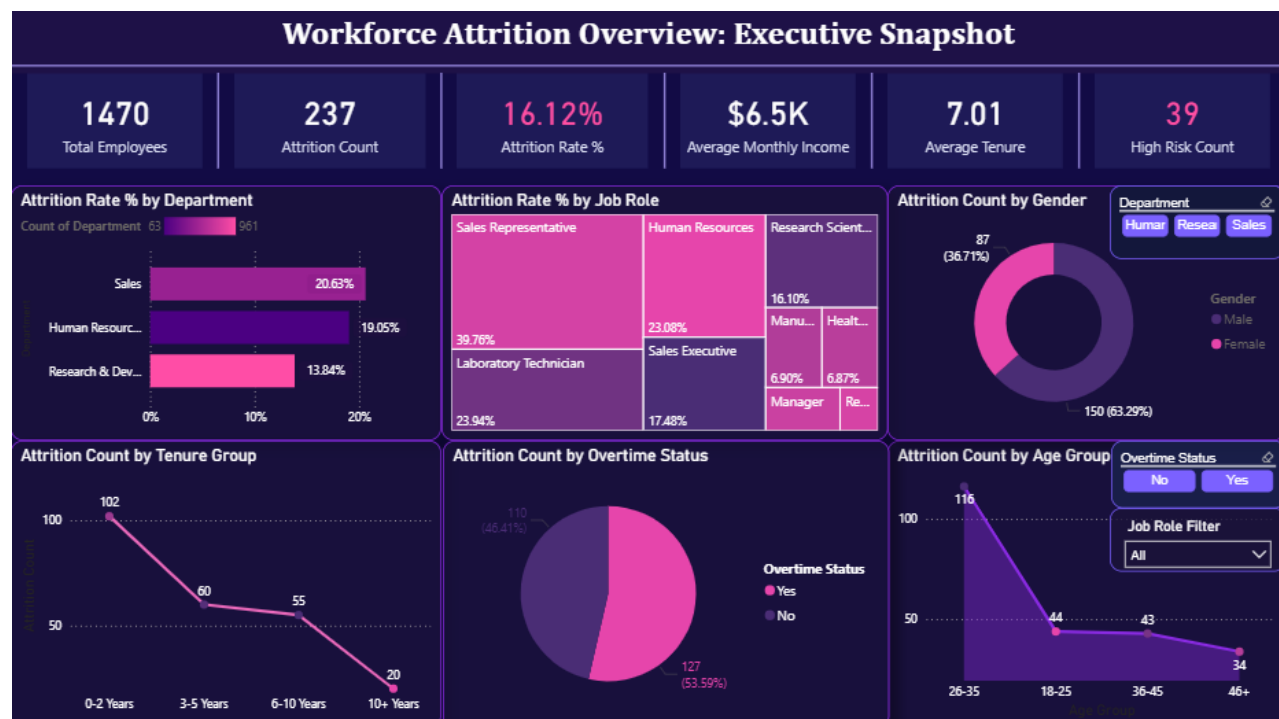
Cleaned and pre-processed IBM HR dataset

4.2 Visualization / Output Results

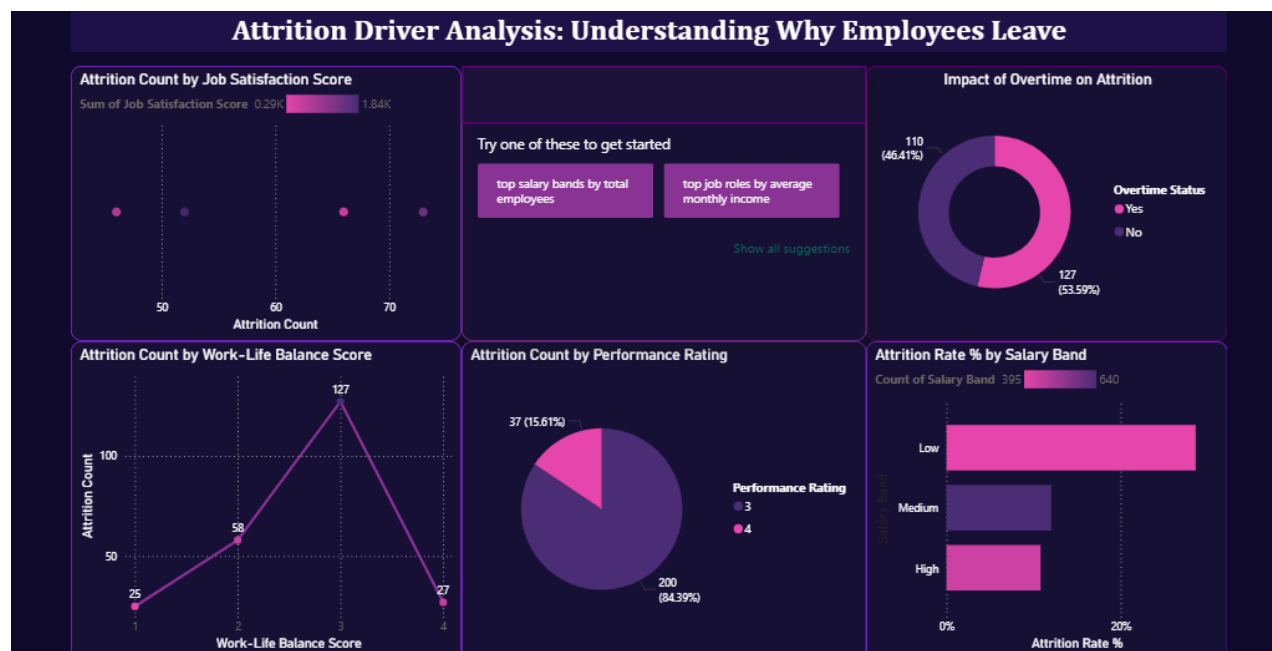
The project results are presented through an interactive Power BI dashboard consisting of multiple visualizations that provide insights into employee attrition and workforce behaviour.

- Dashboard / Chart 1: Executive Overview Dashboard: This dashboard displays key KPIs such as Total Employees, Attrition Count, Attrition Rate, Average Salary, and Average Tenure. It provides a high-level summary of workforce health and attrition trends.

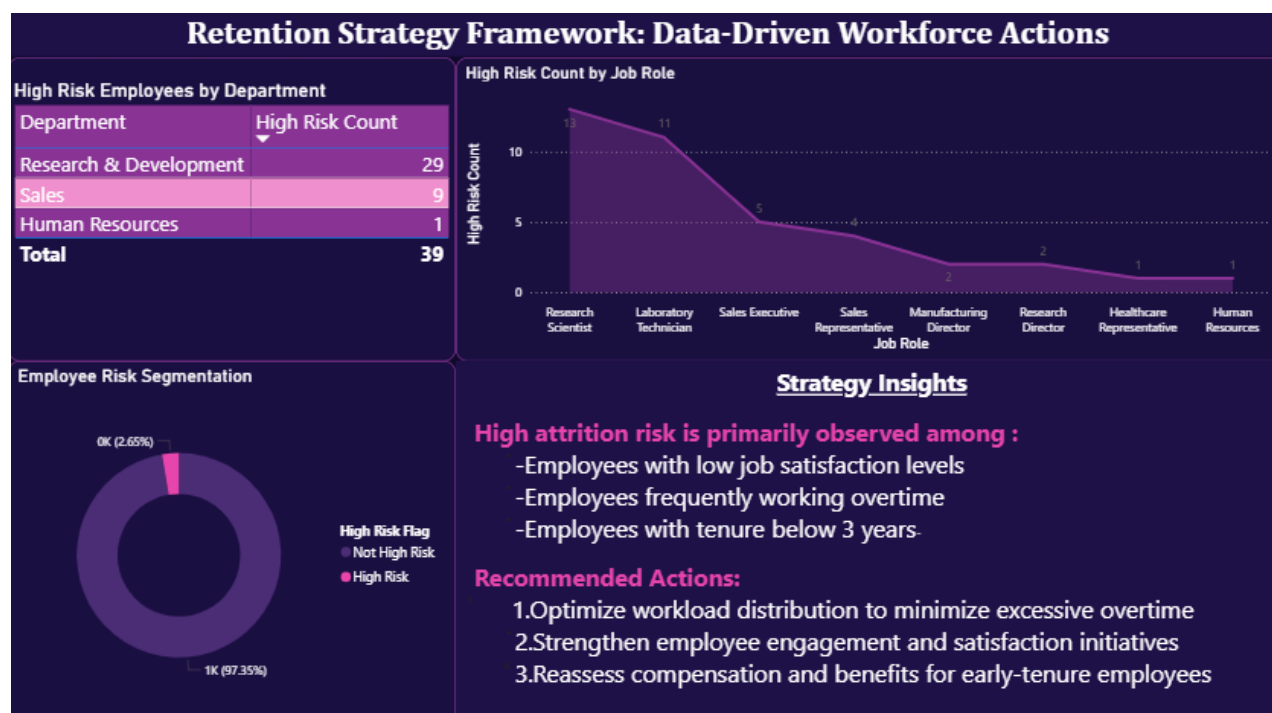
- Dashboard / Chart 2: Attrition Analysis by Department and Job Role: This visualization shows the distribution of attrition across different departments and job roles, helping identify high-risk areas within the organization.
- Dashboard / Chart 3: Attrition Drivers Analysis: This section analyzes factors such as Job Satisfaction, Overtime, Salary Band, and Work-Life Balance to identify key drivers influencing employee attrition.
- Dashboard / Chart 4: Retention Strategy Dashboard: This dashboard highlights high-risk employee segments and provides insights to support proactive retention strategies and workforce planning.



Executive Overview Dashboard (Page 1)



Driver Analysis Dashboard (Page 2)



Retention Strategy Dashboard (Page 3)

4.3 AI Tool Integration Results

- **Tools used:** Zapier, ChatGPT, Azure ML

- **Prompts used:**

- Analyze employee attrition dataset and identify key drivers affecting retention.
- Suggest HR strategies to reduce attrition based on factors like overtime, job satisfaction, and tenure.
- Generate insights and summaries for dashboard reporting in a professional format.

- **Output quality:**

The AI-generated outputs were highly relevant, structured, and easy to interpret. ChatGPT provided clear insights, summaries, and recommendations, while Azure ML generated predictive outputs in the form of attrition probability scores. Zapier ensured accurate and timely data flow for reporting.

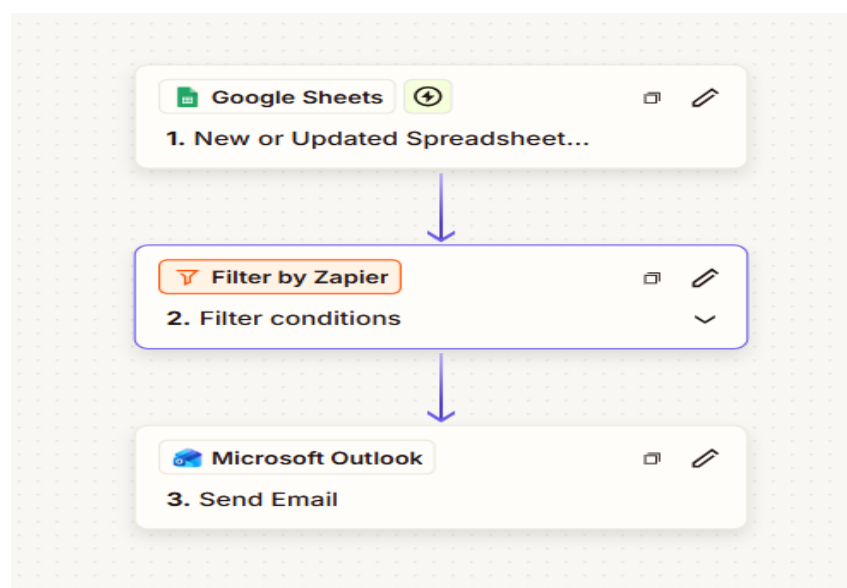
- **How it enhanced the project:**

ChatGPT accelerated data interpretation, insight generation, and content structuring for dashboards. Azure ML enabled predictive analysis by identifying high-risk employees based on key factors such as satisfaction, salary, overtime, and tenure. Zapier automated HR reporting workflows by triggering alerts based on defined conditions, ensuring near real-time updates and reducing manual effort.

- **Automation Workflow (Zapier):**

- Trigger: Google Sheets data update
- Condition: High attrition probability
- Action: Email alert via Gmail/Outlook

Automation Workflow



Automated Output

 **High Attrition Risk Alert – E1470**


Anamika Shah <annexi.mika@gmail.com>
To: anexa7mika@gmail.com





Wed 4/22/2026 6:22 PM

Employee ID: E1470
Prediction: Yes
Risk Level: High
Probability: 85%

 This employee is at high risk of leaving. Immediate HR action recommended.

 Reply

 Forward

Attrition Prediction Algorithms

Microsoft Foundry | Azure Machine Learning

Nirmaan Organization > Workforce Longevity ML > Jobs > Default > attrition-prediction

attrition-prediction    Canceled

Overview Data guardrails **Models + child jobs** Outputs + logs Child jobs

 Refresh  Deploy  Download  Explain model  View generated code  Reset view

Search

Filter

Algorithm name	Responsible AI	AUC weighted	Sampling	Created on
MaxAbsScaler, LightGBM		0.82787	100.00 %	Apr 22, 2026 2:22 PM
StandardScalerWrapper, LightGBM		0.82446	100.00 %	Apr 22, 2026 2:21 PM
MaxAbsScaler, LightGBM		0.82402	100.00 %	Apr 22, 2026 2:28 PM
StandardScalerWrapper, LightGBM		0.82384	100.00 %	Apr 22, 2026 3:35 PM

Page 1 of 4 25/Page

Activate Windows
Go to Settings to activate Windows

Attrition Prediction Table

	A	B	C	D
1	Employee ID	Attrition Prediction	Probability (%)	Risk Level
2	E001	No	43%	Medium
3	E002	No	3%	Low
4	E003	No	82%	Medium
5	E004	No	46%	Medium
6	E005	No	6%	Low
7	E006	No	8%	Low
8	E007	No	47%	Medium
9	E008	No	11%	Low
10	E009	No	10%	Low
11	E010	No	2%	Low
12	E011	No	8%	Low
13	E012	No	8%	Low
14	E013	No	6%	Low
15	E014	No	2%	Low
16	E015	No	84%	Medium
17	E016	No	2%	Low
18	E017	No	29%	Low
19	E018	No	32%	Low

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

- Attrition is higher among employees with low job satisfaction, indicating employee engagement as a critical factor
- Employees working overtime show significantly higher attrition, highlighting workload as a major driver
- Early-tenure employees (0–3 years) exhibit higher attrition rates, indicating challenges in initial employee retention
- Employees in lower salary bands are more likely to leave, emphasizing the importance of competitive compensation
- Specific departments and job roles show concentrated attrition, helping identify high-risk segments within the organization
- Predictive analysis using Azure ML further validated these patterns by identifying employees with a higher probability of attrition
- The integrated system enables rapid identification of high-risk employees, supporting proactive, data-driven, and timely workforce decisions

5.2 Project Links

Share the relevant links to your project work for evaluation.

- GitHub Link of the Project: [https://github.com/anexa7/My_Project/tree/main]
- Video Recording of Project Demonstration: [<https://drive.google.com/file/d/18RTwT9amOEgWITevqfk46RwTOAW8Skhg/view?usp=drivesdk>]
- Video Recording of Implementation (<https://drive.google.com/file/d/1Vy2Cm8f240S6INfZnY3jRfwDy4pBA8t/view?usp=drivesdk>)
- Power BI / Dashboard Link (if applicable): [https://anexa7.github.io/My_Project/]

5.3 Limitations

- The project is based on a static dataset; real-time data integration is simulated using Zapier automation and not implemented as a full enterprise-level system.
- The predictive model developed using Azure ML is a basic implementation and can be further enhanced with advanced machine learning techniques and larger datasets.
- Some AI tool functionalities (such as Copilot) were limited during the project, restricting deeper automation capabilities.
- The dataset used is sample data and may not fully represent real-world organizational complexity.
- The dashboard focuses on key variables and may not include all possible factors influencing employee attrition.

5.4 Future Work

- Enhance automation by integrating advanced workflow tools like Zapier for real-time HR reporting and alerts
- Apply predictive analytics and machine learning models to forecast employee attrition and identify risks in advance
- Enhance the dashboard with automated alerts for high-risk employee segments
- Expand the dataset by including additional factors such as employee engagement surveys and performance feedback
- Develop a scalable and user-friendly interface for non-technical stakeholders
- Create a standardized AI prompt framework for consistent and efficient analysis

5.5 Conclusion

This project successfully addressed the challenge of employee attrition by developing a data-driven employee retention framework using HR analytics and Power BI. The project involved data cleaning, analysis, KPI development, and dashboard visualization to transform raw HR data into meaningful insights.

Key findings revealed that factors such as job satisfaction, overtime, salary, and tenure significantly influence employee attrition. The interactive dashboard developed provides a clear and structured view of workforce trends, enabling better understanding of high-risk employee segments.

All the objectives of the project were successfully achieved, including data preprocessing, analysis, visualization, and the development of a strategic retention framework. This project demonstrates how HR can transition from reactive decision-making to predictive and automated workforce management using analytics, AI, and workflow integration.

This project adds value by providing a practical, scalable, and cost-effective solution that can be applied in real-world organizational settings. It highlights the importance of moving from reactive to proactive approaches in managing employee attrition.

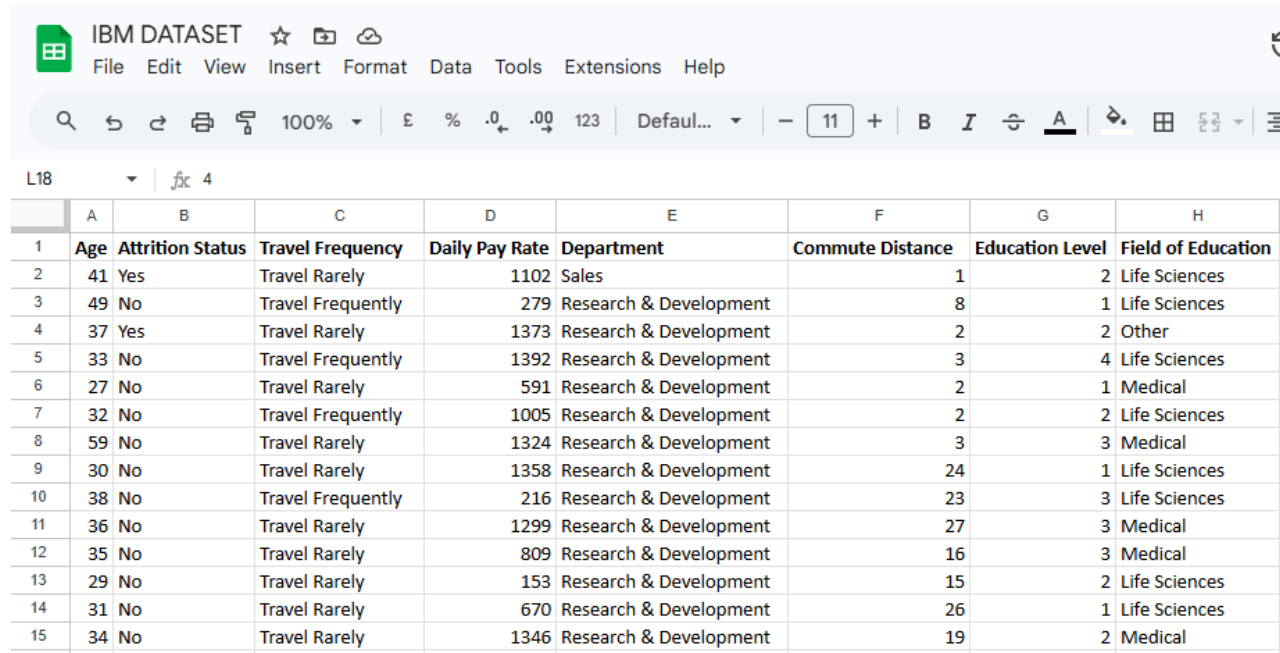
The fellowship experience significantly enhanced practical skills in data analytics, Power BI, and AI-assisted tools, while also improving problem-solving and decision-making abilities. Additionally, the integration of Zapier for automated HR reporting demonstrates how workflow automation can enhance data-driven decision-making and improve operational efficiency.

REFERENCES

- [1] Microsoft Corporation. Microsoft Excel Documentation. Available at: <https://support.microsoft.com/en-us/excel>
- [2] Microsoft Corporation. Power BI Documentation. Available at : <https://learn.microsoft.com/en-us/power-bi/>
- [3] Microsoft Corporation. Microsoft Copilot — AI-Powered Productivity. Available at : <https://copilot.microsoft.com>
- [4] Edunet Foundation. AICW Fellowship Program Guidelines. 2026. Available at: <https://www.edunetfoundation.org>
- [5] LinkedIn Learning. AI and Data Analytics Courses. 2026. Available at: <https://www.linkedin.com/learning>
- [6] Microsoft Corporation. Microsoft Excel Documentation. Available at: <https://support.microsoft.com/en-us/excel>
- [7] Kaggle Inc. Open Datasets. Available at: <https://www.kaggle.com/datasets>
- [8] IBM. (n.d.). *IBM HR Analytics Employee Attrition & Performance Dataset Description*. Available at: <https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>
- [9] Sangu, V. S., Saini, R., Prabakar, S., Hussain, G. K., & Thayumanavar, B. (2024). *HR Analytics: Leveraging Big Data and Artificial Intelligence for Decision-Making in Human Resource Management*. Educational Administration: Theory and Practice. DOI: <https://doi.org/10.53555/kuey.v30i4.2327>
- [10] Talakola, S. (2024). *Enhancing Decision-Making with Data-Driven Insights in Microsoft Power BI*. Essex Journal of AI Ethics and Responsible Innovation. Full Paper: <https://www.ejaei.org/index.php/publication/article/view/61>
- [11] Dhivya, N., & Vithyaramani, G. (2025). *From Data to Decisions: Power BI Visualization for Hiring and Recruitment Analytics*. International Journal for Research in Applied Science & Engineering Technology (IJRASET). DOI: <https://doi.org/10.22214/ijraset.2025.72047>

APPENDICES

Appendix A — Sample Dataset



	A	B	C	D	E	F	G	H
	Age	Attrition Status	Travel Frequency	Daily Pay Rate	Department	Commute Distance	Education Level	Field of Education
1	41	Yes	Travel Rarely	1102	Sales	1	2	Life Sciences
2	49	No	Travel Frequently	279	Research & Development	8	1	Life Sciences
3	37	Yes	Travel Rarely	1373	Research & Development	2	2	Other
4	33	No	Travel Frequently	1392	Research & Development	3	4	Life Sciences
5	27	No	Travel Rarely	591	Research & Development	2	1	Medical
6	32	No	Travel Frequently	1005	Research & Development	2	2	Life Sciences
7	59	No	Travel Rarely	1324	Research & Development	3	3	Medical
8	30	No	Travel Rarely	1358	Research & Development	24	1	Life Sciences
9	38	No	Travel Frequently	216	Research & Development	23	3	Life Sciences
10	36	No	Travel Rarely	1299	Research & Development	27	3	Medical
11	35	No	Travel Rarely	809	Research & Development	16	3	Medical
12	29	No	Travel Rarely	153	Research & Development	15	2	Life Sciences
13	31	No	Travel Rarely	670	Research & Development	26	1	Life Sciences
14	34	No	Travel Rarely	1346	Research & Development	19	2	Medical

Appendix B — Key Excel Formulas / DAX Measures Used

- =COUNTIF(range, criteria) — used to count employees with attrition (“Yes”)
- =IF(condition, value_if_true, value_if_false) — used for categorizing risk levels
- **Power BI DAX:**
 - Total Employees = COUNT(Employee[EmployeeID]) — used in KPI card
 - Attrition Count = CALCULATE(COUNT(Employee[EmployeeID]), Employee[Attrition] = "Yes") — used to measure total attrition
 - Attrition Rate % = DIVIDE([Attrition Count], [Total Employees], 0) — used to calculate attrition percentage
 - Average Monthly Income = AVERAGE(Employee[MonthlyIncome]) — used for income KPI
 - Average Tenure = AVERAGE(Employee[YearsAtCompany]) — used for tenure KPI
 - High Risk Count = CALCULATE(COUNT(Employee[EmployeeID]), Employee[OverTime] = "Yes") — used to identify high-risk employees

Appendix C — AI Prompts Used

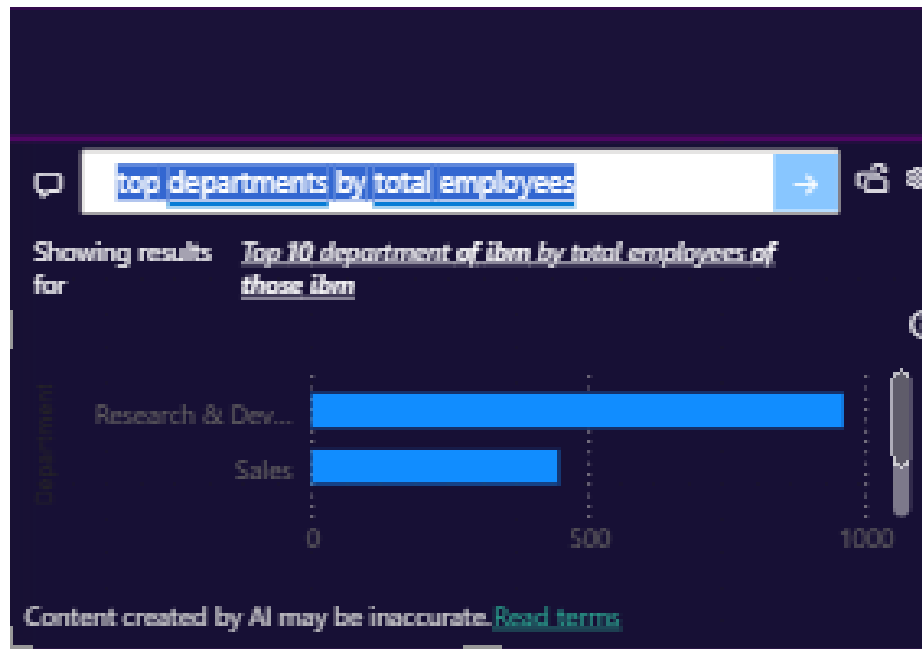
- Prompt 1: “Analyze this employee attrition dataset and identify the key factors influencing employee turnover.”
- Prompt 2: “Generate insights for an HR dashboard explaining attrition trends by job role, department, and tenure.”
- Prompt 3: “Suggest actionable HR strategies to reduce attrition based on factors like overtime, job satisfaction, and salary.”

- Prompt 4: “Create a professional summary of workforce attrition insights for presentation slides.”
- Prompt 5: “Explain patterns and anomalies in employee attrition data and highlight high-risk segments.”

Appendix D — Screenshot Gallery

Attritio...	Attritio...	False_p...	True_pr...	Employ...	Age_orig	Travel ...	Daily P...	Depart...
True	False	0.57473...	0.42526...	E001	41	Travel R...	1102	Sales
False	False	0.97327...	0.02672...	E002	49	Travel Fr...	279	Researc...
True	True	0.18205...	0.81794...	E003	37	Travel R...	1373	Researc...
False	False	0.53944...	0.46055...	E004	33	Travel Fr...	1392	Researc...
False	False	0.94039...	0.05960...	E005	27	Travel R...	591	Researc...
False	False	0.91504...	0.08495...	E006	32	Travel Fr...	1005	Researc...
False	False	0.53384...	0.46615...	E007	59	Travel R...	1324	Researc...

Azure ML Model Prediction Table



Q&A feature in Power BI for dynamic data querying

Strategy Insights

High attrition risk is primarily observed among :

- Employees with low job satisfaction levels
- Employees frequently working overtime
- Employees with tenure below 3 years

Recommended Actions:

- 1.Optimize workload distribution to minimize excessive overtime
- 2.Strengthen employee engagement and satisfaction initiatives
- 3.Reassess compensation and benefits for early-tenure employees

AI-generated insights using Power BI / ChatGPT